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Electrochemical cell

Abstract

An electrochemical cell includes a cell stack including a plurality of unit cells stacked on one another and each being a plate and including an electricity generator and a casing accommodating the electricity generator, and an outer case accommodating the cell stack. The outer case includes at least one rib in a surface area between an outer edge of the outer case and an outer periphery of the cell stack as viewed in a stacking direction of the cell stack. The at least one rib is a portion of the surface area protruding outward.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9203064	12/2014	Lee	N/A	H01M 50/119
2010/0266883	12/2009	Koetting	429/96	H01M 10/6551
2012/0064383	12/2011	Tonomura	429/99	H01M 50/557
2013/0209850	12/2012	Yokoyama	429/99	H01M 10/0481
2018/0212207	12/2017	Mino et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3890045	12/2020	EP	N/A
2000-285967	12/1999	JP	N/A
2010055955	12/2009	JP	N/A
201218866	12/2011	JP	N/A
5287104	12/2012	JP	N/A
2015230812	12/2014	JP	N/A
WO-2011108857	12/2010	WO	H01M 10/0413
2016084273	12/2015	WO	N/A
2020110937	12/2019	WO	N/A

OTHER PUBLICATIONS

Machine Translation of JP-2010055955-A (Year: 2010). cited by examiner

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Background/Summary

RELATED APPLICATIONS

(1) The present application is a National Phase of International Application Number PCT/JP2020/042857, filed Nov. 17, 2020, which claims priority to Japanese Application No. 2019-214401, filed Nov. 27, 2019.

FIELD

(2) The present disclosure relates to an electrochemical cell.

BACKGROUND

(3) A known technique is described in, for example, Patent Literature 1.

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Patent No. 5287104

BRIEF SUMMARY

(5) An electrochemical cell according to an aspect of the present disclosure includes a cell stack including a plurality of unit cells stacked on one another and each being a plate and including an electricity generator and a casing accommodating the electricity generator, and an outer case accommodating the cell stack. The outer case includes at least one rib in a surface area between an outer edge of the outer case and an outer periphery of the cell stack as viewed in a stacking direction of the cell stack. The at least one rib is a portion of the surface area protruding outward.

(6) An electrochemical cell module according to another aspect of the present disclosure includes a plurality of the above electrochemical cells, a current collector electrically connecting the plurality of electrochemical cells to one another, and a housing accommodating the plurality of electrochemical cells.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The objects, features, and advantages of the present disclosure will become more apparent from the following detailed description and the drawings.

(2) FIG. 1 is a plan view of an example electrochemical cell according to an embodiment of the present disclosure.

(3) FIG. 2 is a cross-sectional view of the electrochemical cell taken along line A-A in FIG. 1.

(4) FIG. 3 is a cross-sectional view of the electrochemical cell taken along line B-B in FIG. 1.

(5) FIG. 4 is a plan view of a unit cell included in the electrochemical cell in FIG. 1.

(6) FIG. 5 is a cross-sectional view of the unit cell taken along line C-C in FIG. 4.

(7) FIG. 6 is an exploded perspective view of an example electrochemical cell module according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(8) As an electrochemical cell with the structure that forms the basis of the structure according to one or more embodiments of the present disclosure, a known electrochemical cell includes a cell body functioning as a battery accommodated in an outer case (for example, Patent Literature 1). Such an electrochemical cell has connection terminals connectable to an external device. The connection terminals extending from the inside to the outside of the outer case are connected to the cell body inside the outer case.

(9) In the electrochemical cell with the structure that forms the basis of the structure according to one or more embodiments of the present disclosure, the outer case may easily deform due to, for example, an impact, thus causing misalignment of the cell body accommodated in the outer case. This may cause, for example, deformation of portions connecting the connection terminals to the cell body. The electrochemical cell may thus have lower durability.

(10) An electrochemical cell according to one or more embodiments of the present disclosure will now be described with reference to the drawings.

(11) FIG. 1 is a plan view of an example electrochemical cell according to an embodiment of the present disclosure. FIG. 2 is a cross-sectional view of the electrochemical cell taken along line A-A in FIG. 1. FIG. 3 is a cross-sectional view of the electrochemical cell taken along line B-B in FIG. 1. FIG. 4 is a plan view of a unit cell included in the electrochemical cell in FIG. 1. FIG. 5 is a cross-sectional view of the unit cell taken along line C-C in FIG. 4.

(12) An electrochemical cell **1** according to the present embodiment includes a cell stack **10** and an outer case **20**. The cell stack **10** is a stack of multiple unit cells **11**. The unit cell **11** is a plate. The unit cell **11** is the smallest unit member functioning as a battery in the electrochemical cell **1**.

(13) The unit cell **11** has a first main surface **11a** and a second main surface **11b** opposite to the first main surface **11a**. The unit cell **11** may be, for example, rectangular, square, circular, oval, or in any other shape as viewed in the stacking direction (the lateral direction in FIG. 2, or hereafter simply a first direction) of the cell stack **10**. The unit cell **11** in the present embodiment is, as shown in FIG. 4 for example, substantially rectangular as viewed in the first direction. The unit cell **11** has, for example, a long side of 50 to 500 mm and a short side of 50 to 300 mm as viewed in the first direction. The unit cell **11** has a thickness of, for example, 0.1 to 2 mm in the first direction.

(14) The unit cell **11** includes an electricity generator **12**, a package **13**, a positive electrode terminal **14**, and a negative electrode terminal **15**. The electricity generator **12** charges and discharges through an electrochemical reaction. The electricity generator **12** includes, for example, a positive electrode **12a**, a negative electrode **12b**, and a separator **12c** between the positive electrode **12a** and the negative electrode **12b**. The electricity generator **12** can exchange cations and anions between the positive electrode **12a** and the negative electrode **12b** through the separator **12c**. The electricity generator **12** with the positive electrode **12a** and the negative electrode **12b** electrically connected to an external device can supply electricity to the external device.

(15) The positive electrode **12a** and the negative electrode **12b** are, for example, electrochemically active. The positive electrode **12a** and the negative electrode **12b** may include, for example, an active material and an electrolyte. The electrolyte may be, for example, a solvent containing salt or a solvent mixture containing salt.

(16) The positive electrode **12a** may contain, for example, lithium nickel cobalt aluminum oxide (NCA), lithium manganese oxide spinel (LMO), lithium iron phosphate (LFP), lithium cobalt oxide (LCO), and lithium nickel cobalt manganese oxide (NCM). The positive electrode **12a** may contain, for example, solid-state compounds used for, for example, nickel metal hydride batteries and nickel-cadmium batteries as known to those skilled in the art. The positive electrode **12a** may contain, for example, magnesium (Mg)-doped $\text{LiCoO}_{0.8}\text{Ni}_{0.2}$ and $\text{LiNiO}_{0.8}\text{Co}_{0.2}$.

(17) The negative electrode **12b** may contain, for example, carbon-based materials such as graphite, hard carbon, soft carbon, carbon nanotubes, and graphene. The negative electrode **12b** may contain, for example, titanium oxide such as lithium titanate and titanium dioxide. The negative electrode **12b** may contain, for example, a transition metal compound containing, for example, iron, cobalt, copper, manganese, or nickel.

(18) When the electrochemical cell **1** is a lithium-ion battery, the electrolyte may be, for example, a solvent containing lithium salt. Examples of lithium salt contained in the electrolyte include LiPF_6 , LiBF_4 , LiClO_4 , and LiFSI . Examples of the solvent contained in the electrolyte include propylene carbonate (PC), ethylene carbonate (EC), dimethyl carbonate (DMC), dimethoxyethane (DME), diethyl carbonate (DEC), tetrahydrofuran (THF), and triethylene glycol

dimethyl ether (triglyme).

(19) The separator **12c** reduces the likelihood of a short circuit between the positive electrode **12a** and the negative electrode **12b**. For example, the separator **12c** may have pores for passage of cations and anions. The separator **12c** may be formed from, for example, a porous insulator. Examples of the porous insulator used for the separator **12c** include polyolefin and polyvinyl chloride.

(20) The electricity generator **12** may be, for example, rectangular, square, circular, oval, or in any other shape as viewed in the first direction. In the present embodiment, the electricity generator **12** is, as shown in FIG. 4 for example, rectangular as viewed in the first direction. The electricity generator **12** has, for example, a long side of 50 to 500 mm and a short side of 50 to 300 mm as viewed in the first direction. The electricity generator **12** has a thickness of, for example, 0.1 to 2 mm in the first direction.

(21) In the present embodiment, the multiple unit cells **11** are electrically connected to one another in parallel. This can increase the capacity of the electrochemical cell **1**. The multiple unit cells **11** may be electrically connected to one another in series. This can increase the voltage across the electrochemical cell **1**.

(22) The package **13** electrically insulates the electricity generator **12** from the external environment to protect the electricity generator **12** from the external environment. The package **13** entirely covers and accommodates the electricity generator **12**. The package **13** is, for example, a flat bag. The package **13** may be formed by, for example, welding two laminated films together. The package **13** is formed from, for example, a laminated film shaped into a flat bag. The package **13** may be, for example, rectangular, square, or in any other shape as viewed in the first direction. In the present embodiment, the package **13** is, as shown in FIG. 4 for example, rectangular as viewed in the first direction.

(23) The package **13** includes, for example, an insulator. This can reduce the likelihood of a short circuit between the external environment and the electricity generator **12** through the package **13**, protecting the electricity generator **12** from the external environment. The package **13** includes, for example, a resin material. The resin material may be, for example, polyethylene terephthalate or polyethylene. Thus, the electricity generator **12** is less likely to be in contact with oxygen and moisture contained in the atmosphere, protecting the electricity generator **12** from the external environment.

(24) The package **13** may be, for example, multilayered. The package **13** includes, for example, a thermally adhesive resin material and a heat-resistant resin material. The thermally adhesive resin material melts, for example, at temperatures lower than 150° C. The thermally adhesive resin material may be, for example, polyethylene or polypropylene. The heat-resistant resin material melts, for example, at 150 to 300° C. inclusive. The heat-resistant resin material may be, for example, polyethylene terephthalate or polyethylene naphthalate.

(25) Through the positive electrode terminal **14** and the negative electrode terminal **15**, electricity charged in the electricity generator **12** is output from the package **13**. The positive electrode terminal **14** and the negative electrode terminal **15** extend from the inside to the outside of the package **13**.

(26) The positive electrode terminal **14** is electrically connected to the positive electrode **12a**. The positive electrode terminal **14** is electrically insulated from the negative electrode **12b** and the negative electrode terminal **15**. The positive electrode terminal **14** is formed from, for example, a metal material. The metal material used for the positive electrode terminal **14** may be, for example, aluminum.

(27) The positive electrode terminal **14** includes a first positive electrode terminal portion **14a** inside the package **13**, and a second positive electrode terminal portion **14b** outside the package **13**. The first positive electrode terminal portion **14a** is in contact with the positive electrode **12a**. The first positive electrode terminal portion **14a** is located between the package **13** and the positive

electrode **12a**. The second positive electrode terminal portion **14b** is connected to a connection terminal of the electrochemical cell **1**. The second positive electrode terminal portion **14b** may be, for example, a rectangular or square plate, or in any other shape. In the present embodiment, the second positive electrode terminal portion **14b** is, as shown in FIG. 4 for example, rectangular as viewed in the first direction. The second positive electrode terminal portion **14b** has, for example, a long side of 30 to 100 mm and a short side of 10 to 100 mm as viewed in the first direction. The second positive electrode terminal portion **14b** has a thickness of, for example, 3 to 30 μm in the first direction.

(28) The negative electrode terminal **15** is electrically connected to the negative electrode **12b**. The negative electrode terminal **15** is electrically insulated from the positive electrode **12a** and the positive electrode terminal **14**. The negative electrode terminal **15** is formed from, for example, a metal material. The metal material used for the negative electrode terminal **15** may be, for example, copper.

(29) The negative electrode terminal **15** includes a first negative electrode terminal portion **15a** inside the package **13**, and a second negative electrode terminal portion **15b** outside the package **13**. The first negative electrode terminal portion **15a** is in contact with the negative electrode **12b**. The first negative electrode terminal portion **15a** is located between the package **13** and the negative electrode **12b**. The second negative electrode terminal portion **15b** is connected to the connection terminal of the electrochemical cell **1**. The second negative electrode terminal portion **15b** may be, for example, a rectangular or square plate, or in any other shape. In the present embodiment, the second negative electrode terminal portion **15b** is, as shown in FIG. 4 for example, rectangular as viewed in the first direction. The second negative electrode terminal portion **15b** has, for example, a long side of 30 to 100 mm and a short side of 10 to 100 mm as viewed in the first direction. The second negative electrode terminal portion **15b** has a thickness of, for example, 3 to 30 μm in the first direction.

(30) The second positive electrode terminal portion **14b** and the second negative electrode terminal portion **15b** may, as shown in FIG. 4 for example, extend outward from one side of the package **13** as viewed in the first direction. The second positive electrode terminal portion **14b** and the second negative electrode terminal portion **15b** may extend outward from different sides of the package **13** as viewed in the first direction.

(31) The outer case **20** protects the cell stack **10** from the external environment. The external environment includes, for example, oxygen and moisture in the air. The outer case **20** entirely covers and accommodates the cell stack **10**. The outer case **20** may be, for example, a cylinder, a bag, or in any other shape. The outer case **20** may be formed by, for example, welding two members together into a bag. The outer case **20** may be formed from, for example, a single member shaped into a bag. The outer case **20** may be, for example, rectangular, square, or in any other shape as viewed in the first direction. In the present embodiment, the outer case **20** is, as shown in FIG. 1 for example, rectangular as viewed in the first direction. The outer case **20** has its long-side and short-side directions substantially corresponding to the long-side and short-side directions of the cell stack **10** as viewed in the first direction. The outer case **20** may have, for example, a long side of 50 to 600 mm and a short side of 50 to 400 mm as viewed in the first direction. The outer case **20** has its portion overlapping the cell stack **10** with a thickness of, for example, 50 to 300 μm as viewed in the first direction.

(32) The electrochemical cell **1** includes a connection terminal **30**. Through the connection terminal **30**, the electricity charged in the cell stack **10** is output from the outer case **20**. The connection terminal **30** includes a first connection terminal **31** and a second connection terminal **32**. The first connection terminal **31** and the second connection terminal **32** extend from the inside to the outside of the outer case **20**. The first connection terminal **31** has its portion located inside the outer case **20** and bonded to the multiple positive electrode terminals **14** connected to one another. The second connection terminal **32** has its portion located inside the outer case **20** and bonded to the multiple

negative electrode terminals **15** connected to one another. The first connection terminal **31** and the second connection terminal **32** include, for example, a metal material. Examples of the metal material for the first connection terminal **31** and the second connection terminal **32** include copper and aluminum.

(33) The outer case **20** includes, for example, an insulator. This can reduce the likelihood of a short circuit between the external environment and the cell stack **10** through the outer case **20**, protecting the cell stack **10** from the external environment. The insulator may be, for example, a resin material such as polyethylene terephthalate and polyethylene.

(34) The outer case **20** may be, for example, multilayered. The outer case **20** may be, for example, three-layered. The outer case **20** may include, for example, a first insulating layer, a moisture-proof layer, and a second insulating layer. The moisture-proof layer is located between the first insulating layer and the second insulating layer. The moisture-proof layer may be covered with the first insulating layer and the second insulating layer. The moisture-proof layer may be in direct contact with the first insulating layer and the second insulating layer.

(35) The first insulating layer may be the outermost layer of the three layers in the outer case **20**. The first insulating layer may include a resin material such as polyethylene terephthalate and polyethylene naphthalate. The moisture-proof layer reduces the likelihood that oxygen or moisture penetrating the first insulating layer reaches the second insulating layer. The moisture-proof layer may include a metal material, for example, copper or aluminum. The second insulating layer may include a resin material such as polyethylene and polypropylene.

(36) The outer case **20** may include a liquid layer **21** to transmit external pressure to the unit cells **11**. The liquid layer **21** is located between two adjacent unit cells **11**. The liquid layer **21** may be in direct contact with the two adjacent unit cells **11**. Although the unit cells **11** each may have recesses on its first main surface **11a** and its second main surface **11b**, the liquid layer **21** may be located inside the recesses to allow pressure to be applied uniformly to the two adjacent unit cells **11**. In other words, this structure allows the two adjacent unit cells **11** to perform charging and discharging reactions without varying interface resistances. The unit cells **11** are thus less likely to deteriorate. Each electrochemical cell **1** can thus have a longer service life.

(37) The liquid layer **21** may be located, as shown in FIGS. **2** and **3** for example, between the cell stack **10** and the outer case **20**. Such a cell stack **10** is thus less likely to be misaligned in the outer case **20**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**.

(38) The liquid layer **21** may be, for example, an organic solvent. Examples of the organic solvent used for the liquid layer **21** include ethylene carbonate and γ -butyrolactone. The liquid layer **21** may also be formed from, for example, a flowable low-molecular-weight polymer material such as polyethylene oxide. The liquid layer **21** may also be formed from a silicon-based polymer material such as silicone.

(39) The liquid layer **21** may be formed from, for example, a water-absorbent material such as a water-absorbent polymer. The liquid layer **21** can thus absorb moisture entering the outer case **20** and reduce entry of moisture into the unit cells **11**. Each electrochemical cell **1** can thus have a longer service life. The water-absorbent polymer used for the liquid layer **21** may be, for example, polyacrylonitrile.

(40) The liquid layer **21** may include, for example, an inorganic material such as a porous filler. The liquid layer **21** can thus absorb moisture entering the outer case **20** and reduce entry of moisture into the unit cells **11**. Each electrochemical cell **1** can thus have a longer service life. The porous filler used for the liquid layer **21** may be, for example, zeolite.

(41) The liquid layer **21** may include a metal filler reactable with oxygen and water. Oxygen and water entering the outer case **20** may react with the metal filler and thus are less likely to enter the unit cells **11**. Each electrochemical cell **1** can thus have a longer service life. Examples of the metal

filler used for the liquid layer **21** include iron, copper, and aluminum.

(42) The liquid layer **21** may include a material having higher thermal conductivity than an electrolyte used in the electricity generator **12**. This facilitates transfer of heat generated in the unit cells **11** to the liquid layer **21**. The unit cells **11** are thus less likely to accumulate heat. Each electrochemical cell **1** can thus have a longer service life.

(43) The liquid layer **21** may include a material having higher viscosity than the electrolyte used in the electricity generator **12**. The cell stack **10** is thus less likely to be misaligned in the outer case **20**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**. Each electrochemical cell **1** can thus have a longer service life.

(44) In the electrochemical cell **1** in the embodiment, as shown in FIGS. **1** to **3** for example, the outer case **20** includes at least one rib **20c**. The rib **20c** is located, as shown in FIG. **1** for example, on a surface area **20b** between an outer edge **20a** of the outer case **20** and an outer periphery **10a** of the cell stack **10** in the outer case **20** as viewed in the first direction. The rib **20c** is a portion of the surface area **20b** protruding outward. The rib **20c** may not protrude in the first direction. The rib **20c** may protrude in a direction intersecting with the first direction. The rib **20c** may be located either on a first main surface **20d** or on a second main surface **20e** of the outer case **20**. The rib **20c** may be located both on the first main surface **20d** and on the second main surface **20e** of the outer case **20**.

(45) The outer case **20** including the rib **20c** may be formed, for example, with the method described below. An enclosure to be the outer case **20** as a bag with at least one opening is prepared first. A rib precursor protruding, for example, outward at a predetermined position is formed on the enclosure. The cell stack **10**, the liquid layer **21**, and other components are then arranged inside the enclosure. The enclosure is then evacuated. In this state, the outer case **20** having the rib **20c** is formed by welding the opening of the enclosure.

(46) In the electrochemical cell **1**, the outer case **20** including a rib **20c** is less likely to deform upon, for example, receiving an impact. Thus, the cell stack **10** is less likely to be misaligned in the outer case **20**, and the unit cells **11** are less likely to be misaligned from one another in the cell stack **10**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**, increasing the durability of the electrochemical cell **1**.

(47) The rib **20c** may include, as shown in FIG. **1** for example, a rib **20c1** extending along the outer periphery **10a** of the cell stack **10**. The outer case **20** with the rib **20c1** is further less likely to deform upon, for example, receiving an impact. Thus, the cell stack **10** is further less likely to be misaligned in the outer case **20**, and the unit cells **11** are further less likely to be misaligned from one another in the cell stack **10**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**, increasing the durability of the electrochemical cell **1**.

(48) The rib **20c** may include, as shown in FIG. **1** for example, a rib **20c2** extending along the corner of the outer periphery **10a**. The outer case **20** in the electrochemical cell **1** is susceptible, at its corner of the outer edge **20a**, to an impact during, for example, transportation or assembly of an electrochemical cell module. As shown in FIG. **1** for example, the outer case **20** with the rib **20c2** along the corner of the outer periphery **10a** inward from the corner of the outer edge **20a** is further less likely to deform upon for example, receiving an impact. Thus, the cell stack **10** is further less likely to be misaligned in the outer case **20**, and the unit cells **11** are further less likely to be misaligned from one another in the cell stack **10**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**, increasing the durability of the electrochemical cell **1**.

(49) The ribs **20c2** may be located on the four corners of the outer periphery **10a**. The outer case **20**

is thus further less likely to deform upon, for example, receiving an impact. Thus, the cell stack **10** is further less likely to be misaligned in the outer case **20**, and the unit cells **11** are further less likely to be misaligned from one another in the cell stack **10**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**, increasing the durability of the electrochemical cell **1**.

(50) The rib **20c** may include a rib **20c3** extending from the corner of the outer periphery **10a** of the cell stack **10** to the outer edge **20a** of the outer case **20**. The outer case **20** with the rib **20c3** is further less likely to deform upon, for example, receiving an impact. Thus, the cell stack **10** is further less likely to be misaligned in the outer case **20**, and the unit cells **11** are further less likely to be misaligned from one another in the cell stack **10**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**, increasing the durability of the electrochemical cell **1**.

(51) The rib **20c** may extend fully along the outer periphery **10a**. The outer case **20** with the rib **20c** is further less likely to deform upon, for example, receiving an impact. Thus, the cell stack **10** is further less likely to be misaligned in the outer case **20**, and the unit cells **11** are further less likely to be misaligned from one another in the cell stack **10**. This reduces the likelihood of damage at the joint between the connection terminal **30** and the positive electrode terminals **14** or between the connection terminal **30** and the negative electrode terminals **15**, increasing the durability of the electrochemical cell **1**.

(52) An electrochemical cell module according to one or more embodiments of the present disclosure will now be described. FIG. **6** is an exploded perspective view of an example electrochemical cell module according to an embodiment of the present disclosure.

(53) An electrochemical cell module **3** according to the present embodiment includes multiple electrochemical cells **1**, a current collector **40**, and a housing **50**.

(54) The multiple electrochemical cells **1** are stacked on one another in a predetermined second direction. The multiple electrochemical cells **1** are stacked on one another with their profiles substantially in conformance with one another as viewed in the second direction, forming an electrochemical cell stack **2**. The electrochemical cell stack **2** is, as shown in FIG. **6** for example, substantially rectangular. The multiple electrochemical cells **1** each include the first connection terminal **31** and the second connection terminal **32** protruding from an upper surface **2a** of the electrochemical cell stack **2**.

(55) The current collector **40** electrically connects the multiple electrochemical cells **1** to one another. The current collector **40** may electrically connect the multiple electrochemical cells **1** in parallel. In other words, the current collector **40** may electrically connect the first connection terminals **31** of the multiple electrochemical cells **1** together, and the second connection terminals **32** of the multiple electrochemical cells **1** together. This can increase the capacity of the electrochemical cell module **3**.

(56) To electrically connect the multiple electrochemical cells **1** in series, the current collector **40** may electrically connect the first connection terminals **31** and the second connection terminals **32** of the multiple electrochemical cells **1** to form a series connection. This can increase the voltage across the electrochemical cell module **3**.

(57) The housing **50** accommodates the electrochemical cell stack **2** (multiple electrochemical cells **1**) and protects the electrochemical cell stack **2** from the external environment. The housing **50** may protect the electrochemical cell stack **2** against an external force from the external environment. In the present embodiment, the housing **50** may be, as shown in FIG. **6** for example, a rectangular box open at one surface. The housing **50** may include, for example, a single part shaped into a rectangular prism open at one surface. The housing **50** may also include, for example, two or more parts combined together.

(58) The housing 50 may include, for example, a metal material. Such a housing 50 has higher rigidity and reduces transmission of any external force from the external environment to the electrochemical cell stack 2. The housing 50 thus protects the electrochemical cell stack 2 from the external environment. The metal material used for the housing 50 may be, for example, aluminum or stainless steel. This facilitates transfer of heat generated in the electrochemical cell stack 2 to the housing 50. The electrochemical cell stack 2 is thus less likely to accumulate heat. Each electrochemical cell 1 can thus have a longer service life.

(59) The housing 50 may include, for example, multiple parts. The multiple parts may include, for example, two main surface plates 51, two side plates 52, and a bottom plate 53. The main surface plates 51, the side plates 52, and the bottom plate 53 may include a metal material and a resin material.

(60) The two main surface plates 51 protect end faces 2b and 2c of the electrochemical cell stack 2 in the second direction. The two main surface plates 51 face the respective end faces 2b and 2c of the electrochemical cell stack 2. The main surface plates 51 may be rectangular as viewed in the second direction. In this case, the main surface plates 51 may have, for example, a long side of 200 to 600 mm and a short side of 50 to 300 mm. The main surface plates 51 may have, for example, a thickness of 0.5 to 5 mm.

(61) The main surface plates 51 may include, for example, a metal material. Examples of the metal material used for the main surface plates 51 include aluminum and stainless steel. This facilitates transfer of heat generated in the electrochemical cells 1 outside through the main surface plates 51. Each electrochemical cell 1 can thus have a longer service life.

(62) The main surface plates 51 may include, for example, a resin material. The resin material used for the main surface plates 51 may be, for example, a heat-resistant resin material such as polyethylene terephthalate (PET). Such main surface plates 51 can electrically insulate the electrochemical cells 1 from the external environment, thus reducing the likelihood of a short circuit between the electrochemical cells 1 and the external environment. Each electrochemical cell 1 can thus have a longer service life.

(63) The two side plates 52 protect side surfaces 2d and 2e each connected to the upper surface 2a of the electrochemical cell stack 2 and parallel to the second direction. The two side plates 52 face the respective side surfaces 2d and 2e of the electrochemical cell stack 2.

(64) The side plates 52 may be in contact with at least one of the side surfaces 2d or 2e of the electrochemical cell stack 2. This facilitates transfer of heat generated in the electrochemical cells 1 outside through the side plates 52. Each electrochemical cell 1 can thus have a longer service life. Although at least one of the side plates 52 is in contact with either the side surface 2d or the side surface 2e of the electrochemical cell stack 2, the electrochemical cell module 3 in the present embodiment including the electrochemical cells 1 in the above embodiment can reduce the likelihood of deformation of the outer case 20 and of misalignment of the cell stack 10 in the outer case 20. The electrochemical cells 1 are thus less likely to have lower durability.

(65) The side plates 52 may be rectangular as viewed in a direction perpendicular to the side surfaces 2d and 2e of the electrochemical cell stack 2. In this case, the side plates 52 may have, for example, a long side of 200 to 600 mm and a short side of 50 to 300 mm. The side plates 52 may have, for example, a thickness of 0.5 to 5 mm.

(66) The side plates 52 may include, for example, a metal material. Examples of the metal material used for the side plates 52 include aluminum and stainless steel. This facilitates transfer of heat generated in the electrochemical cells 1 outside through the side plates 52. Each electrochemical cell 1 can thus have a longer service life.

(67) The side plates 52 may include, for example, a resin material. The resin material used for the side plates 52 may be, for example, a heat-resistant resin material such as PET. Such side plates 52 can electrically insulate the electrochemical cells 1 from the external environment, thus reducing the likelihood of a short circuit between the electrochemical cells 1 and the external environment.

Each electrochemical cell **1** can thus have a longer service life.

(68) The bottom plate **53** protects a lower surface **2f** opposite to the upper surface **2a** of the electrochemical cell stack **2**. The bottom plate **53** may be, for example, bent parts of the main surface plates **51** or the side plates **52**.

(69) The bottom plate **53** may be in contact with the lower surface **2f** opposite to the upper surface **2a** of the electrochemical cell stack **2**. This facilitates transfer of heat generated in the electrochemical cells **1** outside through the bottom plate **53**. Each electrochemical cell **1** can thus have a longer service life. Although the bottom plate **53** is in contact with the lower surface **2f** of the electrochemical cell stack **2**, the electrochemical cell module **3** in the present embodiment including the electrochemical cells **1** in the above embodiment can reduce the likelihood of deformation of the outer case **20** and of misalignment of the cell stack **10** in the outer case **20**. The electrochemical cells **1** are thus less likely to have lower durability.

(70) The bottom plate **53** may be rectangular as viewed in a direction perpendicular to the lower surface **2f** of the electrochemical cell stack **2**. In this case, the bottom plate **53** may have, for example, a long side of 200 to 600 mm and a short side of 50 to 300 mm. The bottom plate **53** may have, for example, a thickness of 0.5 to 5 mm.

(71) The bottom plate **53** may include, for example, a metal material. Examples of the metal material used for the bottom plate **53** include aluminum and stainless steel. This facilitates transfer of heat generated in the electrochemical cells **1** outside through the bottom plate **53**. Each electrochemical cell **1** can thus have a longer service life.

(72) The bottom plate **53** may include, for example, a resin material. The resin material used for the bottom plate **53** may be, for example, a heat-resistant resin material such as PET. Such a bottom plate **53** can electrically insulate the electrochemical cells **1** from the external environment, thus reducing the likelihood of a short circuit between the electrochemical cells **1** and the external environment. Each electrochemical cell **1** can thus have a longer service life.

(73) The electrochemical cell stack **2** may be held with their end faces **2b** and **2c** in the second direction pressed in the housing **50**. The electrochemical cell stack **2** may be pressed and held by pressure plates **54** and elastic members **55**.

(74) The pressure plates **54** may be formed from, for example, a metal material. This facilitates transfer of heat generated in the electrochemical cells **1** outside. Each electrochemical cell **1** can thus have a longer service life. Examples of the metal material used for the pressure plates **54** include aluminum and stainless steel.

(75) The pressure plates **54** may also be formed from, for example, an insulating resin material. Such pressure plates **54** can electrically insulate the electrochemical cells **1** from the external environment, thus reducing the likelihood of a short circuit between the electrochemical cells **1** and the external environment. The resin material used for the pressure plates **54** may be, for example, a thermosetting resin such as an epoxy resin, a phenolic resin, or a melamine resin.

(76) The pressure plates **54** may also be formed from, for example, both a resin material and a metal material. The resin material may be used for portions of the pressure plates **54** in contact with the electrochemical cell stack **2**. Such portions can electrically insulate the electrochemical cells **1** from the pressure plates **54**, thus reducing the likelihood of a short circuit between the electrochemical cells **1** and the external environment. The pressure plates **54** including a metal material are less likely to be damaged.

(77) The elastic members **55** are located between the pressure plates **54** and the main surface plates **51** of the housing **50**. The elastic members **55** apply pressure to the pressure plates **54** to apply pressure to the electrochemical cells **1**. The elastic members **55** may be, for example, springs. The springs may be, for example, helical. The springs may be, for example, bent plates. The springs may include, for example, a metal material. Examples of the metal material used for the springs include steel and stainless steel. The helical springs may have, for example, a diameter of 5 to 50 mm, a length of 10 to 50 mm, and a pitch of 1 to 10 mm.

(78) The elastic members 55 may be, for example, a rubber material shaped into plates. The rubber plates may have, for example, the same shape as the pressure plates 54. The rubber material may be, for example, natural rubber or synthetic rubber.

(79) The electrochemical cell stack 2 may have their end faces 2b and 2c in the second direction receiving pressure applied by the housing 50. The housing 50 may apply pressure to the end faces 2b and 2c of the electrochemical cell stack 2 with, for example, their main surface plates 51 fastened to the side plates 52 with screws to press the end faces 2b and 2c of the electrochemical cell stack 2.

(80) The electrochemical cell module 3 including the electrochemical cells 1 can achieve an electrochemical cell module with higher durability. In addition, the electrochemical cell module 3 including the housing 50 with higher rigidity may reduce transmission of any external force from the external environment to the electrochemical cells 1. Such an electrochemical cell module can have still higher durability.

(81) Electrochemical Cell System

(82) An electrochemical cell system according to an embodiment of the present disclosure includes the electrochemical cell module in the above embodiment, and a controller to control the electrochemical cell module. The controller may be a control integrated circuit (IC) incorporated in the electrochemical cell module to protect a battery from overcharge and over-discharge. The control IC may include a protection circuit as a program.

(83) In response to the battery voltage exceeding a preset fully charged battery voltage, the protection circuit forcibly stops the flow of a charging current. In response to a battery voltage during discharge being less than a preset dischargeable voltage, the protection circuit forcibly stops discharging the current. Such sudden power shutdown due to the operation of the protection circuit can greatly affect facilities (power receivers) receiving power from the electrochemical cell module. To avoid such a situation, in response to the battery voltage decreasing to the dischargeable voltage, a user may receive warning information such as a message stating “Low battery. Charge now.” on a display, with a sound, or in any other manner. In response to the warning signal, the user stores data into, for example, a memory and then stops the facility.

(84) The protection circuit implements a charging sequence including, for example, pre-charging, quick charging or constant-current charging, constant-voltage charging, and determining that the battery is fully charged in this order. In the pre-charging, a small amount of current flows in the battery first. In response to the battery reaching a constant voltage, charging with a constant current (constant-current charging) is performed. In this manner, constant-voltage constant-current (CVCC) control shifts the constant current to the constant voltage. In response to the battery voltage increasing substantially to a fully charged voltage, the protection circuit performs charging at a constant voltage. During the constant-voltage charging, the charging current naturally decreases as the internal voltage of the battery increases. In response to the current value decreasing to a constant current value, the protection circuit determines that the charging is complete and thus stops charging.

(85) The present disclosure may be implemented in the following forms.

(86) An electrochemical cell according to one or more embodiments of the present disclosure includes a cell stack including a plurality of unit cells stacked on one another and each being a plate and including an electricity generator and a casing accommodating the electricity generator, and an outer case accommodating the cell stack. The outer case includes at least one rib in a surface area between an outer edge of the outer case and an outer periphery of the cell stack as viewed in a stacking direction of the cell stack. The at least one rib is a portion of the surface area protruding outward.

(87) An electrochemical cell module according to one or more embodiments of the present disclosure includes a plurality of the above electrochemical cells, a current collector electrically connecting the plurality of electrochemical cells to one another, and a housing accommodating the

plurality of electrochemical cells.

(88) An electrochemical cell system according to one or more embodiments of the present disclosure includes at least one electrochemical cell module, and a controller that controls the electrochemical cell module.

(89) The electrochemical cells according to one or more aspects of the present disclosure can reduce the likelihood of deformation of the outer case and misalignment of the cell stack in the outer case. The electrochemical cells can thus have higher durability. The electrochemical cell module according to one or more aspects of the present disclosure including the electrochemical cells described above and the electrochemical cell system including the electrochemical cell module can have higher durability.

(90) Although embodiments of the present disclosure have been described in detail, the present disclosure is not limited to the embodiments described above, and may be changed or modified in various manners without departing from the spirit and scope of the present disclosure. The components described in the above embodiments may be entirely or partially combined as appropriate unless any contradiction arises.

REFERENCE SIGNS LIST

(91) **1** electrochemical cell **2** electrochemical cell stack **2a** upper surface **2b, 2c** end face **2d, 2e** side surface **2f** lower surface **3** electrochemical cell module **10** cell stack **10a** outer periphery **11** unit cell **11a** first main surface **11b** second main surface **12** electricity generator **12a** positive electrode **12b** negative electrode **12c** separator **13** package **14** positive electrode terminal **14a** first positive electrode terminal portion **14b** second positive electrode terminal portion **15** negative electrode terminal **15a** first negative electrode terminal portion **15b** second negative electrode terminal portion **20** outer case **20a** outer edge **20b** surface area **20c, 20c1, 20c2, 20c3** rib **20d** first main surface **20e** second main surface **21** liquid layer **30** connection terminal **31** first connection terminal **32** second connection terminal **40** current collector **50** housing **51** main surface plate **52** side plate **53** bottom plate **54** pressure plate **55** elastic member

Claims

1. An electrochemical cell, comprising: a cell stack including a plurality of unit cells stacked on one another in a stacking direction of the cell stack, each of the plurality of unit cells having a plate shape and including an electricity generator and a casing accommodating the electricity generator; and an outer case accommodating the cell stack, the outer case including an outer layer having a folded portion located in a surface area of the outer layer between an outer edge of the outer case and an outer periphery of the cell stack in a direction perpendicular to the stacking direction, the folded portion of the outer layer defining at least one rib protruding outward in the stacking direction from the surface area of the outer layer.

2. The electrochemical cell according to claim 1, wherein the at least one rib includes a rib extending along the outer periphery of the cell stack.

3. The electrochemical cell according to claim 1, wherein the cell stack is rectangular as viewed in the stacking direction, and the at least one rib includes a rib extending along a corner of the outer periphery of the cell stack.

4. The electrochemical cell according to claim 3, wherein the at least one rib includes four ribs, and each of the four ribs extends along a corresponding corner of four corners of the outer periphery of the cell stack.

5. The electrochemical cell according to claim 3, wherein the at least one rib includes a further rib extending from the corner of the outer periphery of the cell stack towards the outer edge of the outer case.

6. The electrochemical cell according to claim 1, wherein the at least one rib includes a rib extending along an entirety of the outer periphery of the cell stack.

7. The electrochemical cell according to claim 1, wherein the electricity generator includes a positive electrode, a negative electrode, and a separator located between the positive electrode and the negative electrode.
 8. The electrochemical cell according to claim 5, wherein the further rib extends from the corner of the outer periphery of the cell stack towards a corner of the outer edge of the outer case.
 9. The electrochemical cell according to claim 1, wherein the outer case is a bag having first and second members welded to each other.
 10. The electrochemical cell according to claim 1, wherein the outer case is a single member shaped in a bag.
 11. The electrochemical cell according to claim 1, wherein the outer layer of the outer case is an insulating layer including a resin material.
 12. The electrochemical cell according to claim 1, wherein the outer layer of the outer case has a first main surface and a second main surface opposite to the first main surface in the stacking direction, and the at least one rib is located on at least one of the first main surface or the second main surface.
 13. The electrochemical cell according to claim 12, wherein the at least one rib is located on both of the first main surface and the second main surface.
 14. An electrochemical cell module, comprising: a plurality of electrochemical cells; a current collector electrically connecting the plurality of electrochemical cells to one another; and a housing accommodating the plurality of electrochemical cells, wherein each of the plurality of electrochemical cells includes: a cell stack including a plurality of unit cells stacked on one another in a stacking direction of the cell stack, each of the plurality of unit cells having a plate shape and including an electricity generator and a casing accommodating the electricity generator; and an outer case accommodating the cell stack, the outer case including an outer layer having a folded portion located in a surface area of the outer layer between an outer edge of the outer case and an outer periphery of the cell stack in a direction perpendicular to the stacking direction, the folded portion of the outer layer defining at least one rib protruding outward in the stacking direction from the surface area of the outer layer.
 15. An electrochemical cell system, comprising: at least one electrochemical cell module; and a controller configured to control the at least one electrochemical cell module, wherein the at least one electrochemical cell module includes: a plurality of electrochemical cells; a current collector electrically connecting the plurality of electrochemical cells to one another; and a housing accommodating the plurality of electrochemical cells, each of the plurality of electrochemical cells includes: a cell stack including a plurality of unit cells stacked on one another in a stacking direction of the cell stack, each of the plurality of unit cells having a plate shape and including an electricity generator and a casing accommodating the electricity generator; and an outer case accommodating the cell stack, the outer case including an outer layer having a folded portion located in a surface area of the outer layer between an outer edge of the outer case and an outer periphery of the cell stack in a direction perpendicular to the stacking direction, the folded portion of the outer layer defining at least one rib protruding outward in the stacking direction from the surface area of the outer layer.
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